

Chapter 6: We Distribute, Yet Things Multiply (Algebraic Expressions and Identities)

Worksheet 6: Algebraic Identities and Multiplication of Polynomials

Learning Objective: Students will multiply monomials, binomials, and trinomials; apply standard algebraic identities $(a + b)^2$, $(a - b)^2$, $(a + b)(a - b)$, $(x + a)(x + b)$; factorise using identities; and solve problems using algebraic expressions.

Worked Example

Expand using identity: $(3x + 2y)^2$

Solution: $(a + b)^2 = a^2 + 2ab + b^2$

$$(3x + 2y)^2 = 9x^2 + 12xy + 4y^2$$

Find 103^2 using identity: $(100 + 3)^2 = 10000 + 600 + 9 = 10609$

Questions 1–5: Easy / Recall

Q1. State the identity for $(a + b)^2$.

Q2. Expand: $(x + 5)^2$

Q3. Expand: $(2y - 3)^2$

Q4. Expand: $(x + 4)(x - 4)$

Q5. Multiply: $3x \times 4x^2$

Questions 6–10: Basic Application

Q6. Expand: $(2a + 3b)(2a - 3b)$

Q7. Expand: $(x + 7)(x + 3)$

Q8. Use identity to find: 99^2

Q9. Use identity to find: 102×98

Q10. Multiply: $(x + 2y)(x^2 - 2xy + 4y^2)$. (Hint: Use $(a + b)(a^2 - ab + b^2) = a^3 + b^3$.)

Questions 11–15: Practice and Skill Building

Q11. Factorise: $25x^2 - 16y^2$

Q12. Factorise: $x^2 + 10x + 25$

Q13. Factorise: $4a^2 - 12ab + 9b^2$

Q14. Expand and simplify: $(a + b + c)^2$ and verify for $a = 1, b = 2, c = 3$.

Q15. If $x + \frac{1}{x} = 5$, find $x^2 + \frac{1}{x^2}$ and $x^4 + \frac{1}{x^4}$.

Questions 16–18: Higher Order Thinking

Q16. If $a - b = 4$ and $ab = 21$, find $a^2 + b^2$ and $(a + b)^2$.

Q17. Prove geometrically that $(a + b)^2 = a^2 + 2ab + b^2$ using a square diagram. (Teacher Verification Required)

Q18. Use the identity $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ to expand $(2x + y)^3$.

Questions 19–20: Real-Life Applications

Q19. A square garden of side $(x + 5)$ m has a smaller square of side $(x - 2)$ m inside it. Find the area of the border region between them. Simplify using identities. If $x = 8$, find the actual area.

Q20. A rectangle has length $(3a + 2)$ cm and breadth $(2a - 1)$ cm. Find its area as a polynomial. If $a = 4$, find the actual dimensions and verify the area.

Reflection Question: Explain in your own words how algebraic identities act as shortcuts. Give one example where using an identity saves significant calculation effort.